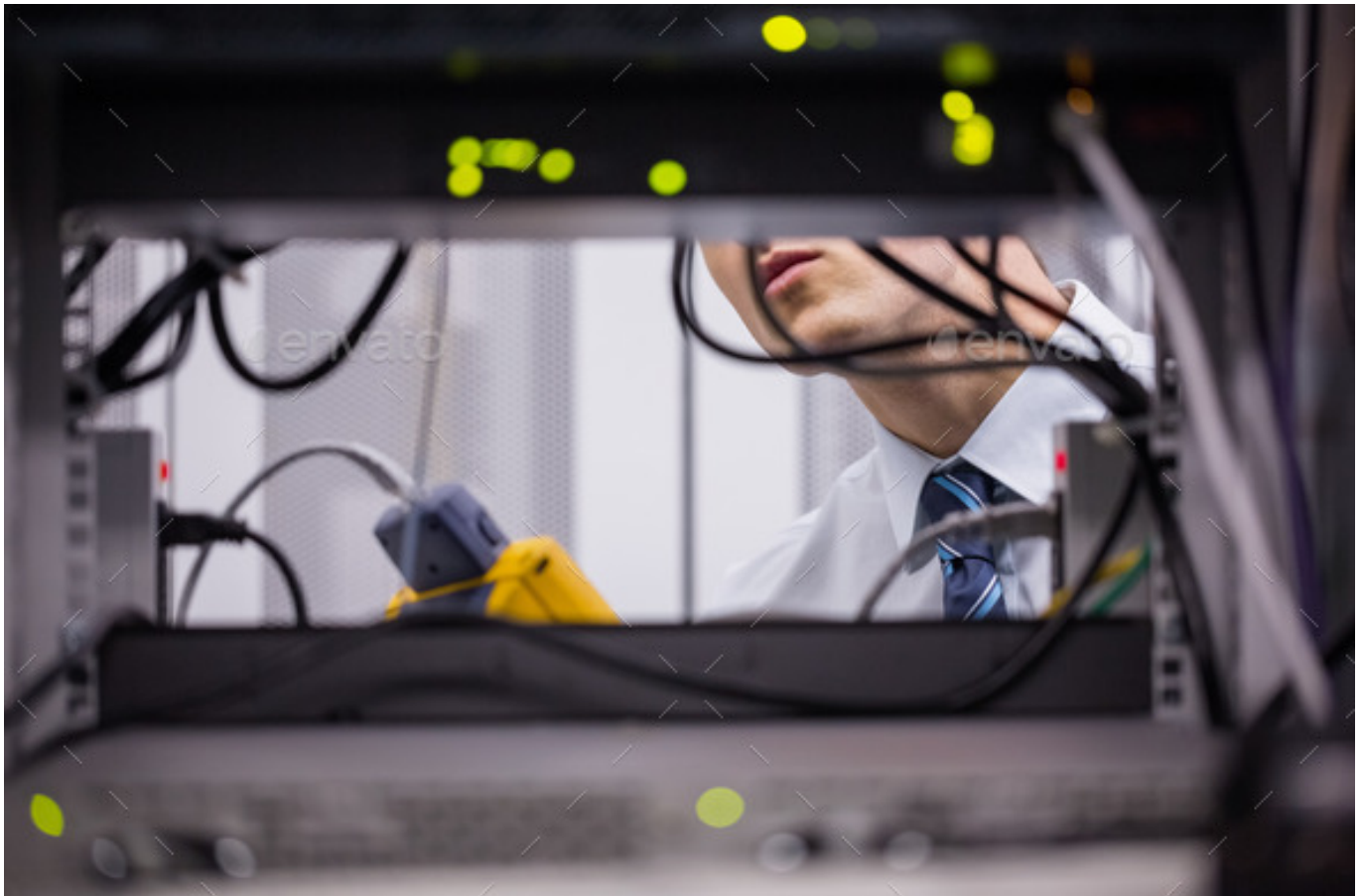


## Qualification Pack



# Digital Cable Technician - Access

QP Code: ELE/Q8106

Version: 4.0

NSQF Level: 4

Electronics Sector Skills Council of India || 155, 2nd Floor, ESC House Okhla Industrial Area-Phase 3  
New Delhi- 110020 || email:standards@essc-india.org

## Qualification Pack

## Contents

|                                                                                                     |    |
|-----------------------------------------------------------------------------------------------------|----|
| ELE/Q8106: Digital Cable Technician - Access .....                                                  | 3  |
| <i>Brief Job Description</i> .....                                                                  | 3  |
| Applicable National Occupational Standards (NOS) .....                                              | 3  |
| <i>Compulsory NOS</i> .....                                                                         | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                                     | 3  |
| ELE/N8116: Lay truck or sub truck cables install and connectorize as per designed route .....       | 5  |
| ELE/N8117: Carry out installation of power inserter & balancing and commissioning of amplifier .... | 11 |
| ELE/N8118: Install Set Top Box at customers premise .....                                           | 16 |
| ELE/N8119: Cater to customer complaints for troubleshooting and maintenance of set top box .....    | 23 |
| ELE/N9905: Work effectively at the workplace .....                                                  | 27 |
| ELE/N1002: Apply health and safety practices at the workplace .....                                 | 35 |
| DGT/VSQ/N0102: Employability Skills (60 Hours) .....                                                | 43 |
| ELE/N8120: Lay, install and carry out commissioning for multi play network .....                    | 51 |
| Assessment Guidelines and Weightage .....                                                           | 59 |
| <i>Assessment Guidelines</i> .....                                                                  | 59 |
| <i>Assessment Weightage</i> .....                                                                   | 60 |
| Acronyms .....                                                                                      | 61 |
| Glossary .....                                                                                      | 62 |

## Qualification Pack

### ELE/Q8106: Digital Cable Technician - Access

#### Brief Job Description

The person is responsible for installation, maintenance and repair of cable television and set-top box at premises. The technicians work on the trunk line and sub-trunk line, and inspect cable lines, lay cables under or on top of the building and repair cable faults

#### Personal Attributes

The individual need to have a high level of manual dexterity/capability and need to have basic knowledge in math and science. They also need to be good communicators and listeners, and should be able to solve problems independently.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [ELE/N8116: Lay truck or sub truck cables install and connectorize as per designed route](#)
2. [ELE/N8117: Carry out installation of power inserter & balancing and commissioning of amplifier](#)
3. [ELE/N8118: Install Set Top Box at customers premise](#)
4. [ELE/N8119: Cater to customer complaints for troubleshooting and maintenance of set top box](#)
5. [ELE/N9905: Work effectively at the workplace](#)
6. [ELE/N1002: Apply health and safety practices at the workplace](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
8. [ELE/N8120: Lay, install and carry out commissioning for multi play network](#)

#### Qualification Pack (QP) Parameters

|                   |                                |
|-------------------|--------------------------------|
| <b>Sector</b>     | Electronics                    |
| <b>Sub-Sector</b> | Communication and Broadcasting |
| <b>Occupation</b> | After Sales Service            |
| <b>Country</b>    | India                          |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NSQF Level</b>                                         | 4                                                                                                                                                                                                                                                                                            |
| <b>Credits</b>                                            | 20                                                                                                                                                                                                                                                                                           |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/NIL                                                                                                                                                                                                                                                                                 |
| <b>Minimum Educational Qualification &amp; Experience</b> | 8th grade pass with 2 years of NTC (plus 2 year of NAC/relevant Experience)<br>OR<br>10th grade pass (plus 2 year of NTC/NAC/relevant experience)<br>OR<br>12th Class<br>OR<br>Certificate-NSQF (Level 3 in Domain of Electrical/Electronic/Mechanical/relevant ) with 2 Years of experience |
| <b>Minimum Level of Education for Training in School</b>  | 10th Class                                                                                                                                                                                                                                                                                   |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                           |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                     |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                           |
| <b>Next Review Date</b>                                   | 27/01/2027                                                                                                                                                                                                                                                                                   |
| <b>NSQC Approval Date</b>                                 | 27/01/2022                                                                                                                                                                                                                                                                                   |
| <b>Version</b>                                            | 4.0                                                                                                                                                                                                                                                                                          |
| <b>Reference code on NQR</b>                              | 2022/EHW/ESSC/06675                                                                                                                                                                                                                                                                          |
| <b>NQR Version</b>                                        | 1.0                                                                                                                                                                                                                                                                                          |

### Remarks:

NA

## Qualification Pack

### ELE/N8116: Lay truck or sub truck cables install and connectorize as per designed route

#### Description

This OS unit is about laying coaxial cables, carrying out installation and connectorization of actives and passives as per the designed route

#### Scope

The scope covers the following :

- Lay truck or sub truck cables, install and connectorize as per designed route
- Carry out installation and connectorization of actives and passives

#### Elements and Performance Criteria

##### *Lay truck or sub truck cables install and connectorize as per designed route*

To be competent, the user/individual on the job must be able to:

- PC1.** prepare a plan for laying cables for single way/two way with reverse path as per the designed route provided by the supervisor
- PC2.** identify the various types of cables Types: sub-trunk, access cable and fiber optics
- PC3.** calculate the height and length of the pole/tree before the laying the cables
- PC4.** create a requirement list and procure the materials
- PC5.** test the line using a line tester
- PC6.** ensure to lay the cables on a low-tension line
- PC7.** support the trunk/sub-trunk cable lines evenly about the circumference with messenger wires without crushing the cable
- PC8.** clip the cables using clipping wire at a distance of 1 meter throughout the trunk/subtrunk line
- PC9.** keep the cable away from heating vents and water heaters
- PC10.** maintain a minimum height of 12-16 feet for trunks
- PC11.** carry out laying of cables as per the designed route like from pole to pole or tree to tree, etc. by climbing up the ladder safely
- PC12.** adhere to all safety measures while working on electrical poles Safety measures: Wear safety belts, shock proof shoes, helmets, goggles, and gloves
- PC13.** keep a loop of 3-5 meters cable on every pole after 80 meters
- PC14.** bend the cable as per manufacturers bending specification to avoid shorts, cable's impedance and signal loss

##### *Carry out installation and connectorization of actives and passives*

To be competent, the user/individual on the job must be able to:

- PC15.** carry out proper fitting of the enclosure box for housing (actives/passives)
- PC16.** keep a small cable loop at the input and output of the amplifier
- PC17.** check the box have provision for locking

## Qualification Pack

- PC18.** inspect that the box is painted with proper LCO code
- PC19.** ensure the RF coaxial cable connectors have a characteristic impedance matching that of the cable
- PC20.** check proper connectorization is established by using standard connectors and right kinds of tools
- PC21.** connectorize the trunk cables, input and output of the amplifiers and install passives (Tap off/Splitters) as per the design diagram
- PC22.** create testing points at the input and output
- PC23.** ensure proper torque is given to all the connectors

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on: customer care
- KU2.** companys code of conduct
- KU3.** organisation culture and typical customer profile
- KU4.** companys reporting structure
- KU5.** companys documentation policy
- KU6.** unit losses of the different coaxial cables
- KU7.** various parts of coaxial cables Parts: Centre conductor, insulating dielectric, outer conductor, and outer protecting jacket or sheath
- KU8.** different types of cores used in coaxial cables Types: Copper core and steel with copper cladding
- KU9.** various types of support wires Types: Single strand and multi-strand
- KU10.** loop creation and bending process of cables
- KU11.** various tools used for laying and connectorization of cables Tools: Line tester, cable preparation tools, cutter, pliers, multi-meter, radio frequency meter, screw driver, etc.
- KU12.** connectorization process for actives and passives
- KU13.** proper fitting methods for amplifiers and splitters
- KU14.** etiquettes to be followed at customers premises
- KU15.** precautions to be taken while handling field calls and dealing with customers
- KU16.** relevant reference sheets, manuals and documents to carry in the field

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, maintenance records, customer feedback form, etc.
- GS2.** note problems on job sheet and details of work done
- GS3.** read network design drawing and correctly interpret different legends used in drawing
- GS4.** read warnings, instructions and other text material on product labels, components, etc
- GS5.** read job sheets or work orders

## Qualification Pack

- GS6.** read product and model serial numbers and interpret details such as make
- GS7.** receive and ask for clarifications from supervisor on the job requirement
- GS8.** listen carefully to the customer
- GS9.** communicate in local/English language
- GS10.** educate and inform customers about available connection plans, offers and cost of service
- GS11.** educate on precautions to be taken in order to avoid recurrence of problem
- GS12.** follow standard operating procedures while making decisions
- GS13.** take approval from supervisor in case the decision has to be made for exceptions
- GS14.** work with supervisor and co-workers to achieve smooth workflow
- GS15.** work with superiors and co-workers to share knowledge and learning
- GS16.** maintain personal grooming
- GS17.** be polite, patient and courteous under all circumstances with all types of customers
- GS18.** decide on the spot on whether interaction of customer with superior is necessary or not
- GS19.** maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering customer premises without permission
- GS20.** put customer at ease and generate customers confidence
- GS21.** seek inputs to assess the problems
- GS22.** interpret accurately drawings, wiring and job specifications/instructions
- GS23.** improve work processes

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Lay trunk or sub trunk cables install and connectorize as per designed route</i>                                                                              | <b>26</b>    | <b>38</b>       | -             | -          |
| <b>PC1.</b> prepare a plan for laying cables for single way/two way with reverse path as per the designed route provided by the supervisor                       | 2            | 2               | -             | -          |
| <b>PC2.</b> identify the various types of cables Types: sub-trunk, access cable and fiber optics                                                                 | 2            | 2               | -             | -          |
| <b>PC3.</b> calculate the height and length of the pole/tree before the laying the cables                                                                        | 2            | 3               | -             | -          |
| <b>PC4.</b> create a requirement list and procure the materials                                                                                                  | 2            | 3               | -             | -          |
| <b>PC5.</b> test the line using a line tester                                                                                                                    | 1            | 2               | -             | -          |
| <b>PC6.</b> ensure to lay the cables on a low-tension line                                                                                                       | 2            | 3               | -             | -          |
| <b>PC7.</b> support the trunk/sub-trunk cable lines evenly about the circumference with messenger wires without crushing the cable                               | 2            | 3               | -             | -          |
| <b>PC8.</b> clip the cables using clipping wire at a distance of 1 meter throughout the trunk/subtrunk line                                                      | 1            | 2               | -             | -          |
| <b>PC9.</b> keep the cable away from heating vents and water heaters                                                                                             | 1            | 2               | -             | -          |
| <b>PC10.</b> maintain a minimum height of 12-16 feet for trunks                                                                                                  | 1            | 2               | -             | -          |
| <b>PC11.</b> carry out laying of cables as per the designed route like from pole to pole or tree to tree, etc. by climbing up the ladder safely                  | 4            | 5               | -             | -          |
| <b>PC12.</b> adhere to all safety measures while working on electrical poles Safety measures: Wear safety belts, shock proof shoes, helmets, goggles, and gloves | 2            | 3               | -             | -          |
| <b>PC13.</b> keep a loop of 3-5 meters cable on every pole after 80 meters                                                                                       | 2            | 3               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> bend the cable as per manufacturers bending specification to avoid shorts, cable's impedance and signal loss                         | 2            | 3               | -             | -          |
| <i>Carry out installation and connectorization of actives and passives</i>                                                                        | <b>14</b>    | <b>22</b>       | -             | -          |
| <b>PC15.</b> carry out proper fitting of the enclosure box for housing (actives/passives)                                                         | 2            | 3               | -             | -          |
| <b>PC16.</b> keep a small cable loop at the input and output of the amplifier                                                                     | 2            | 3               | -             | -          |
| <b>PC17.</b> check the box have provision for locking                                                                                             | 2            | 3               | -             | -          |
| <b>PC18.</b> inspect that the box is painted with proper LCO code                                                                                 | 2            | 3               | -             | -          |
| <b>PC19.</b> ensure the RF coaxial cable connectors have a characteristic impedance matching that of the cable                                    | 1            | 2               | -             | -          |
| <b>PC20.</b> check proper connectorization is established by using standard connectors and right kinds of tools                                   | 1            | 2               | -             | -          |
| <b>PC21.</b> connectorize the trunk cables, input and output of the amplifiers and install passives (Tap off/Splitters) as per the design diagram | 2            | 2               | -             | -          |
| <b>PC22.</b> create testing points at the input and output                                                                                        | 1            | 2               | -             | -          |
| <b>PC23.</b> ensure proper torque is given to all the connectors                                                                                  | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                                                  | <b>40</b>    | <b>60</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                              |
|----------------------------|------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ELE/N8116                                                                    |
| <b>NOS Name</b>            | Lay truck or sub truck cables install and connectorize as per designed route |
| <b>Sector</b>              | Electronics                                                                  |
| <b>Sub-Sector</b>          | Communication and Broadcasting                                               |
| <b>Occupation</b>          | After Sales Service                                                          |
| <b>NSQF Level</b>          | 4                                                                            |
| <b>Credits</b>             | 4                                                                            |
| <b>Version</b>             | 2.0                                                                          |
| <b>Last Reviewed Date</b>  | NA                                                                           |
| <b>Next Review Date</b>    | 27/01/2027                                                                   |
| <b>NSQC Clearance Date</b> | 27/01/2022                                                                   |

## Qualification Pack

### ELE/N8117: Carry out installation of power inserter & balancing and commissioning of amplifier

#### Description

This OS unit is about carrying out installation of power inserter and balancing and commissioning of amplifier

#### Scope

The scope covers the following :

- Install power inserter
- Carry out balancing and commissioning of amplifier

#### Elements and Performance Criteria

##### *Install power inserter*

To be competent, the user/individual on the job must be able to:

- PC1.** carry all required tools and test equipment before installing the power inserter
- PC2.** ensure 90 Volt AC /50 Hertz power is inserted to the main trunk lines for powering the trunk amplifiers from the remote location
- PC3.** check the availability of power at all the input points
- PC4.** ensure proper earthing of the line as per the standard grounding parameter
- PC5.** feed the radio frequency (RF) signal from the combiner to the trunk line
- PC6.** check and ensure the input of the trunk amplifier is as per level provided by design team

##### *Carry out balancing and commissioning of amplifier*

To be competent, the user/individual on the job must be able to:

- PC7.** set the amplifier at different stages and balance the output maintaining a slope as specified by the design team
- PC8.** balance the amplifiers for various paths as per amplifier levels provided by the design team various paths: Forward and reverse
- PC9.** record all the levels and maintain a datasheet for future reference
- PC10.** check the name of the nodes and amplifiers are painted on the enclosure boxes as per the area code and box serial number
- PC11.** carry out comprehensive testing for all the cascade amplifiers
- PC12.** measure signal-to-noise ratio (SNR) levels and quality parameters for all amplifiers in coordination with the backend engineering team

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

## Qualification Pack

- KU1.** company's policies on: Incentives, delivery standards, and personnel management, call closure
- KU2.** company's sales, installation and after sales support policy
- KU3.** importance of the individual's role in the workflow
- KU4.** reporting structure
- KU5.** company's policy on products warranty and other terms and conditions
- KU6.** methods to handle 90 V AC
- KU7.** functions of power inserter
- KU8.** frequency spectrum used for community antenna television
- KU9.** process to set-up amplifier using input stage, mid-stage, and output stage
- KU10.** attenuation at different frequency levels for proper slope control
- KU11.** instrument use for balancing amplifier, cable analyser

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, maintenance records, customer feedback form, etc.
- GS2.** note problems on job sheet and details of work done
- GS3.** read warnings, instructions and other text material on product labels, components, etc
- GS4.** read job sheets or work orders
- GS5.** read product and model serial numbers and interpret details such as make
- GS6.** receive and ask for clarifications from supervisor on the job requirement
- GS7.** listen carefully to the customer
- GS8.** communicate in local/English language
- GS9.** educate and inform customers about available connection plans, offers and cost of service
- GS10.** educate on precautions to be taken in order to avoid recurrence of problem
- GS11.** follow standard operating procedures while making decisions
- GS12.** take approval from supervisor in case the decision has to be made for exceptions
- GS13.** work with supervisor and co-workers to achieve smooth workflow
- GS14.** work with superiors and co-workers to share knowledge and learning
- GS15.** maintain personal grooming
- GS16.** be polite, patient and courteous under all circumstances with all types of customers
- GS17.** decide on the spot on whether interaction of customer with superior is necessary or not
- GS18.** maintain proper etiquette including the appropriate physical distance with
- GS19.** customer during conversation, not entering customers premises without permission, etc
- GS20.** put customer at ease and generate customers confidence
- GS21.** seek inputs to assess the problems
- GS22.** interpret accurately drawings, wiring and job specifications/instructions
- GS23.** improve work processes

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Install power inserter</i>                                                                                                                       | <b>14</b>    | <b>22</b>       | -             | -          |
| <b>PC1.</b> carry all required tools and test equipment before installing the power inserter                                                        | 1            | 2               | -             | -          |
| <b>PC2.</b> ensure 90 Volt AC /50 Hertz power is inserted to the main trunk lines for powering the trunk amplifiers from the remote location        | 2            | 2               | -             | -          |
| <b>PC3.</b> check the availability of power at all the input points                                                                                 | 2            | 2               | -             | -          |
| <b>PC4.</b> ensure proper earthing of the line as per the standard grounding parameter                                                              | 3            | 2               | -             | -          |
| <b>PC5.</b> feed the radio frequency (RF) signal from the combiner to the trunk line                                                                | 3            | 7               | -             | -          |
| <b>PC6.</b> check and ensure the input of the trunk amplifier is as per level provided by design team                                               | 3            | 7               | -             | -          |
| <i>Carry out balancing and commissioning of amplifier</i>                                                                                           | <b>26</b>    | <b>38</b>       | -             | -          |
| <b>PC7.</b> set the amplifier at different stages and balance the output maintaining a slope as specified by the design team                        | 5            | 7               | -             | -          |
| <b>PC8.</b> balance the amplifiers for various paths as per amplifier levels provided by the design team various paths: Forward and reverse         | 5            | 7               | -             | -          |
| <b>PC9.</b> record all the levels and maintain a datasheet for future reference                                                                     | 3            | 5               | -             | -          |
| <b>PC10.</b> check the name of the nodes and amplifiers are painted on the enclosure boxes as per the area code and box serial number               | 3            | 5               | -             | -          |
| <b>PC11.</b> carry out comprehensive testing for all the cascade amplifiers                                                                         | 5            | 7               | -             | -          |
| <b>PC12.</b> measure signal-to-noise ratio (SNR) levels and quality parameters for all amplifiers in coordination with the backend engineering team | 5            | 7               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------|--------------|-----------------|---------------|------------|
| NOS Total                        | 40           | 60              | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ELE/N8117                                                                           |
| <b>NOS Name</b>            | Carry out installation of power inserter & balancing and commissioning of amplifier |
| <b>Sector</b>              | Electronics                                                                         |
| <b>Sub-Sector</b>          | Communication and Broadcasting                                                      |
| <b>Occupation</b>          | After Sales Service                                                                 |
| <b>NSQF Level</b>          | 4                                                                                   |
| <b>Credits</b>             | 4                                                                                   |
| <b>Version</b>             | 2.0                                                                                 |
| <b>Last Reviewed Date</b>  | NA                                                                                  |
| <b>Next Review Date</b>    | 27/01/2027                                                                          |
| <b>NSQC Clearance Date</b> | 27/01/2022                                                                          |

## Qualification Pack

### ELE/N8118: Install Set Top Box at customers premise

#### Description

This OS unit is about undertaking pre-installation and installation process of Set Top Box (STB) at customers premise.

#### Scope

The scope covers the following :

- Undertake pre- installation process
- Remove packaging and check accessories
- Install Set Top Box (STB)
- Complete installation documentation
- Interact with customer in a professional manner

#### Elements and Performance Criteria

##### *Undertake pre- installation process*

To be competent, the user/individual on the job must be able to:

- PC1.** visit the customers premise before carrying out the installation
- PC2.** interact with the customer to understand the requirement
- PC3.** check that the location meets structural requirements such as distance from the poles/tree, power points, etc
- PC4.** educate the customer about the installation procedure and different subscription plans
- PC5.** seek appointment for installation

##### *Remove packaging and check accessories*

To be competent, the user/individual on the job must be able to:

- PC6.** ensure to carry sealed pack of set top box from the local cable operator
- PC7.** remove packaging of the set top box in front of the customer
- PC8.** check for all supporting accessories in the pack
- PC9.** carry the tools and fitments required for the installation
- PC10.** dispose the materials as per company's norms

##### *Install Set Top Box (STB)*

To be competent, the user/individual on the job must be able to:

- PC11.** draw drop cables as per customer's requirement inside the customer's premises from the tap poles by clipping at desired cable length
- PC12.** measure the threshold input available at the customer's premises
- PC13.** identify the power point to run the set top box at the customer 's premise
- PC14.** check proper connectorization at both ends of the drop cable
- PC15.** switch on and configure the set top box on all technical parameters Parameters: Carrier to noise ratio, forward error correction, symbol rate, etc.
- PC16.** activate the set top box from the backend team



## Qualification Pack

- PC17.** ensure proper input signal level is established at the set top box (STB) input point
- PC18.** connect the set top box to the television set with composite cables/ high-definition multiple insertion
- PC19.** check for all available channels as per customers requirement on the set top box
- PC20.** explain various preventive maintenance procedures to the customers, to keep the connection running
- PC21.** ensure the work site is cleaned after the work is completed and remove all the wastes from the site

### *Complete installation documentation*

To be competent, the user/individual on the job must be able to:

- PC22.** fill in customer acknowledgement form
- PC23.** seek customers signature
- PC24.** complete other documentation for recording completion of installation
- PC25.** call customer care and inform about job completion

### *Interact with customer in a professional manner*

To be competent, the user/individual on the job must be able to:

- PC26.** be courteous and professional over the call as well as at the merchants premises
- PC27.** be smartly dressed
- PC28.** reach the venue at least prior 5 mins the appointment time
- PC29.** greet the customer and introduce self
- PC30.** conduct a demonstration of how to use set top box in a pace and manner that the customer can follow comfortably
- PC31.** attend to and answer customer queries and doubts
- PC32.** leave the premises by thanking and bidding good bye to the customer in a courteous manner after the work is completed to their satisfaction

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on: Incentives, delivery standards, and personnel management, call closure
- KU2.** companys sales, installation and after sales support policy
- KU3.** importance of the individuals role in the workflow
- KU4.** reporting structure
- KU5.** companys policy on products warranty and other terms and conditions
- KU6.** installation site requirements
- KU7.** waste disposal procedures
- KU8.** tools and equipment for installation Tools: Compression tools, cable preparation tools, tools bag, RF/ dB meter, attenuator/terminator, plier, spanner set, screw driver set, hammer small, line tester, etc.
- KU9.** drop cable losses
- KU10.** input requirement for set top box

## Qualification Pack

- KU11.** configuration and functions of set top box
- KU12.** methods for connecting television and set top box
- KU13.** different television functions like input cables, television modes like digital/analog input, etc.
- KU14.** installation and activation process for set top box
- KU15.** safety rules, policies and procedures
- KU16.** various quality standards

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, maintenance records, customer feedback form, etc.
- GS2.** note problems on job sheet and details of work done
- GS3.** read network design drawing and correctly interpret different legends used in drawing
- GS4.** read job sheets or work orders
- GS5.** read product and model serial numbers and interpret details such as make
- GS6.** receive and ask for clarifications from supervisor on the job requirement
- GS7.** listen carefully to the customer
- GS8.** communicate in local/English language
- GS9.** educate and inform customers about available connection plans, offers and cost of service
- GS10.** educate on precautions to be taken in order to avoid recurrence of problem
- GS11.** follow standard operating procedures while making decisions
- GS12.** take approval from supervisor in case the decision has to be made for exceptions
- GS13.** work with supervisor and co-workers to achieve smooth workflow
- GS14.** work with superiors and co-workers to share knowledge and learning
- GS15.** maintain personal grooming
- GS16.** be polite, patient and courteous under all circumstances with all types of customers
- GS17.** decide on the spot on whether interaction of customer with superior is necessary or not
- GS18.** maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering customer premises without permission
- GS19.** seek inputs to assess the problems
- GS20.** interpret accurately drawings, wiring and job specifications/instructions
- GS21.** improve work processes

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Undertake pre- installation process</i>                                                                                                        | <b>8</b>     | <b>8</b>        | -             | -          |
| <b>PC1.</b> visit the customers premise before carrying out the installation                                                                      | 2            | 1               | -             | -          |
| <b>PC2.</b> interact with the customer to understand the requirement                                                                              | 2            | 1               | -             | -          |
| <b>PC3.</b> check that the location meets structural requirements such as distance from the poles/tree, power points, etc                         | 2            | 3               | -             | -          |
| <b>PC4.</b> educate the customer about the installation procedure and different subscription plans                                                | 1            | 2               | -             | -          |
| <b>PC5.</b> seek appointment for installation                                                                                                     | 1            | 1               | -             | -          |
| <i>Remove packaging and check accessories</i>                                                                                                     | <b>8</b>     | <b>11</b>       | -             | -          |
| <b>PC6.</b> ensure to carry sealed pack of set top box from the local cable operator                                                              | 1            | 1               | -             | -          |
| <b>PC7.</b> remove packaging of the set top box in front of the customer                                                                          | 1            | 1               | -             | -          |
| <b>PC8.</b> check for all supporting accessories in the pack                                                                                      | 2            | 3               | -             | -          |
| <b>PC9.</b> carry the tools and fitments required for the installation                                                                            | 2            | 3               | -             | -          |
| <b>PC10.</b> dispose the materials as per company's norms                                                                                         | 2            | 3               | -             | -          |
| <i>Install Set Top Box (STB)</i>                                                                                                                  | <b>12</b>    | <b>24</b>       | -             | -          |
| <b>PC11.</b> draw drop cables as per customer's requirement inside the customer's premises from the tap poles by clipping at desired cable length | 2            | 5               | -             | -          |
| <b>PC12.</b> measure the threshold input available at the customer's premises                                                                     | 1            | 2               | -             | -          |
| <b>PC13.</b> identify the power point to run the set top box at the customer 's premise                                                           | 1            | 2               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC14.</b> check proper connectorization at both ends of the drop cable                                                                                        | 1            | 2               | -             | -          |
| <b>PC15.</b> switch on and configure the set top box on all technical parameters Parameters: Carrier to noise ratio, forward error correction, symbol rate, etc. | 1            | 2               | -             | -          |
| <b>PC16.</b> activate the set top box from the backend team                                                                                                      | 1            | 2               | -             | -          |
| <b>PC17.</b> ensure proper input signal level is established at the set top box (STB) input point                                                                | 1            | 2               | -             | -          |
| <b>PC18.</b> connect the set top box to the television set with composite cables/ high-definition multiple insertion                                             | 1            | 2               | -             | -          |
| <b>PC19.</b> check for all available channels as per customers requirement on the set top box                                                                    | 1            | 2               | -             | -          |
| <b>PC20.</b> explain various preventive maintenance procedures to the customers, to keep the connection running                                                  | 1            | 2               | -             | -          |
| <b>PC21.</b> ensure the work site is cleaned after the work is completed and remove all the wastes from the site                                                 | 1            | 1               | -             | -          |
| <i>Complete installation documentation</i>                                                                                                                       | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC22.</b> fill in customer acknowledgement form                                                                                                               | 1            | 1               | -             | -          |
| <b>PC23.</b> seek customers signature                                                                                                                            | 1            | 1               | -             | -          |
| <b>PC24.</b> complete other documentation for recording completion of installation                                                                               | 1            | 2               | -             | -          |
| <b>PC25.</b> call customer care and inform about job completion                                                                                                  | 1            | 2               | -             | -          |
| <i>Interact with customer in a professional manner</i>                                                                                                           | <b>8</b>     | <b>11</b>       | -             | -          |
| <b>PC26.</b> be courteous and professional over the call as well as at the merchants premises                                                                    | 1            | 2               | -             | -          |
| <b>PC27.</b> be smartly dressed                                                                                                                                  | 1            | 2               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC28.</b> reach the venue at least prior 5 mins the appointment time                                                                                  | 1            | 1               | -             | -          |
| <b>PC29.</b> greet the customer and introduce self                                                                                                       | 1            | 1               | -             | -          |
| <b>PC30.</b> conduct a demonstration of how to use set top box in a pace and manner that the customer can follow comfortably                             | 1            | 2               | -             | -          |
| <b>PC31.</b> attend to and answer customer queries and doubts                                                                                            | 1            | 2               | -             | -          |
| <b>PC32.</b> leave the premises by thanking and bidding good bye to the customer in a courteous manner after the work is completed to their satisfaction | 2            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                                                                         | <b>40</b>    | <b>60</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                          |
|----------------------------|------------------------------------------|
| <b>NOS Code</b>            | ELE/N8118                                |
| <b>NOS Name</b>            | Install Set Top Box at customers premise |
| <b>Sector</b>              | Electronics                              |
| <b>Sub-Sector</b>          | Communication and Broadcasting           |
| <b>Occupation</b>          | After Sales Service                      |
| <b>NSQF Level</b>          | 4                                        |
| <b>Credits</b>             | 3                                        |
| <b>Version</b>             | 2.0                                      |
| <b>Last Reviewed Date</b>  | NA                                       |
| <b>Next Review Date</b>    | 27/01/2027                               |
| <b>NSQC Clearance Date</b> | 27/01/2022                               |

## Qualification Pack

### ELE/N8119: Cater to customer complaints for troubleshooting and maintenance of set top box

#### Description

This OS unit about catering to the customer complaints by troubleshooting and maintenance of the set top box

#### Scope

The scope covers the following :

- Carry out routine check-up and maintenance of the trunk line
- Cater to customer complaints timely

#### Elements and Performance Criteria

##### *Carry out routine check-up and maintenance of the trunk line*

To be competent, the user/individual on the job must be able to:

- PC1.** ensure routine check-up of the trunk line
- PC2.** maintain the trunk and amplifier periodically for its input and output signal variation
- PC3.** inspect the cable for any cut or damage
- PC4.** check the availability of power supply to all the trunk amplifiers are within 60-90 volt
- PC5.** remove packaging of set top box carefully and check accessories before installation at the customers premise

##### *Cater to customer complaints timely*

To be competent, the user/individual on the job must be able to:

- PC6.** diagnose the fault in the unit as per customers complaint and initial inspection
- PC7.** cater to customer's complaints at the earliest and troubleshoot problem timely
- PC8.** check the drop cable from the tap/split port for its functioning patch cord
- PC9.** inspect the connectors at both ends
- PC10.** check the power supply of set top box /television /router/ optical network unit (ONU)
- PC11.** verify customer's authorization for establishing network signal as per cable/ data plan
- PC12.** check all the output connectivity from the optical network unit
- PC13.** collect monthly subscription from the customers on a regular basis

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on: Incentives, delivery standards and personnel management and customer service standards
- KU2.** reporting and documentation processes
- KU3.** reporting structure

## Qualification Pack

- KU4.** output connectivity of optical network unit
- KU5.** standards configuring parameters for set top box
- KU6.** customer feedback process
- KU7.** documentation procedure

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, maintenance records, customer feedback form, etc.
- GS2.** note problems on job sheet and details of work done
- GS3.** read network design drawing and correctly interpret different legends used in drawing
- GS4.** read job sheets or work orders
- GS5.** read product and model serial numbers and interpret details such as make
- GS6.** receive and ask for clarifications from supervisor on the job requirement
- GS7.** listen carefully to the customer
- GS8.** communicate in local/English language
- GS9.** educate and inform customers about available connection plans, offers and cost of service
- GS10.** educate on precautions to be taken in order to avoid recurrence of problem
- GS11.** follow standard operating procedures while making decisions
- GS12.** take approval from supervisor in case the decision has to be made for exceptions
- GS13.** work with supervisor and co-workers to achieve smooth workflow
- GS14.** work with superiors and co-workers to share knowledge and learning
- GS15.** maintain personal grooming
- GS16.** be polite, patient and courteous under all circumstances with all types of customers
- GS17.** decide on the spot on whether interaction of customer with superior is necessary or not
- GS18.** maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering customer premises without permission
- GS19.** seek inputs to assess the problems
- GS20.** interpret accurately drawings, wiring and job specifications/instructions
- GS21.** improve work processes



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Carry out routine check-up and maintenance of the trunk line</i>                                                      | <b>10</b>    | <b>21</b>       | -             | -          |
| <b>PC1.</b> ensure routine check-up of the trunk line                                                                    | 2            | 4               | -             | -          |
| <b>PC2.</b> maintain the trunk and amplifier periodically for its input and output signal variation                      | 2            | 7               | -             | -          |
| <b>PC3.</b> inspect the cable for any cut or damage                                                                      | 2            | 4               | -             | -          |
| <b>PC4.</b> check the availability of power supply to all the trunk amplifiers are within 60-90 volt                     | 2            | 3               | -             | -          |
| <b>PC5.</b> remove packaging of set top box carefully and check accessories before installation at the customers premise | 2            | 3               | -             | -          |
| <i>Cater to customer complaints timely</i>                                                                               | <b>20</b>    | <b>49</b>       | -             | -          |
| <b>PC6.</b> diagnose the fault in the unit as per customers complaint and initial inspection                             | 3            | 7               | -             | -          |
| <b>PC7.</b> cater to customer's complaints at the earliest and troubleshoot problem timely                               | 3            | 7               | -             | -          |
| <b>PC8.</b> check the drop cable from the tap/split port for its functioning patch cord                                  | 3            | 7               | -             | -          |
| <b>PC9.</b> inspect the connectors at both ends                                                                          | 3            | 7               | -             | -          |
| <b>PC10.</b> check the power supply of set top box /television /router/ optical network unit (ONU)                       | 3            | 7               | -             | -          |
| <b>PC11.</b> verify customer's authorization for establishing network signal as per cable/ data plan                     | 1            | 3               | -             | -          |
| <b>PC12.</b> check all the output connectivity from the optical network unit                                             | 3            | 7               | -             | -          |
| <b>PC13.</b> collect monthly subscription from the customers on a regular basis                                          | 1            | 4               | -             | -          |
| <b>NOS Total</b>                                                                                                         | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                 |
|----------------------------|---------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ELE/N8119                                                                       |
| <b>NOS Name</b>            | Cater to customer complaints for troubleshooting and maintenance of set top box |
| <b>Sector</b>              | Electronics                                                                     |
| <b>Sub-Sector</b>          | Communication and Broadcasting                                                  |
| <b>Occupation</b>          | After Sales Service-C&B                                                         |
| <b>NSQF Level</b>          | 4                                                                               |
| <b>Credits</b>             | 2                                                                               |
| <b>Version</b>             | 2.0                                                                             |
| <b>Last Reviewed Date</b>  | NA                                                                              |
| <b>Next Review Date</b>    | 27/01/2027                                                                      |
| <b>NSQC Clearance Date</b> | 27/01/2022                                                                      |

## Qualification Pack

### ELE/N9905: Work effectively at the workplace

#### Description

This unit is about the communicating and managing work effectively at the workplace as well as taking measures to enhance own competence and working in a disciplined and ethical manner.

#### Scope

The scope covers the following :

- Communicate effectively at the workplace
- Work effectively
- Maintain and enhance professional competence
- Work in a disciplined and ethical manner
- Uphold social diversity at the workplace

#### Elements and Performance Criteria

##### *Communicate effectively at the workplace*

To be competent, the user/individual on the job must be able to:

- PC1.** exchange information and instruction with colleagues, and seek clarifications and feedback as necessary
- PC2.** assist colleagues where required
- PC3.** follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person)
- PC4.** document and share all relevant information with stakeholders in agreed formats and as per agreed timelines

##### *Work effectively*

To be competent, the user/individual on the job must be able to:

- PC5.** identify and obtain clarity regarding organisational, team and own goals and targets
- PC6.** prioritise and plan work in order to achieve goals and targets
- PC7.** monitor own and team performance as per agreed plan
- PC8.** complete duties accurately, systematically and within required timeframes
- PC9.** express emotions appropriately at the workplace and manage own response to heightened emotions
- PC10.** maintain orderliness and cleanliness in the work area

##### *Maintain and enhance professional competence*

To be competent, the user/individual on the job must be able to:

- PC11.** identify own strengths and weaknesses in relation to goals and targets
- PC12.** adapt self, service, or product to meet success criteria
- PC13.** seek and select opportunities for continuous professional development
- PC14.** formulate a professional development plan to enhance capabilities

## Qualification Pack

- PC15.** build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations
- PC16.** examine developments and trends in field of work and their potential impact on work
- PC17.** take feedback from peers, supervisors and clients to improve own performance and practices

*Work in a disciplined and ethical manner*

To be competent, the user/individual on the job must be able to:

- PC18.** perform tasks as per workplace standards, organisational policies and legislative requirements
- PC19.** display appropriate professional appearance at the workplace and adhere to the organisational dress code
- PC20.** demonstrate responsible and disciplined behaviour at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behaviour at all times, adopting environment- friendly practices, etc.
- PC21.** identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution
- PC22.** protect the rights of the client and organisation when delivering services
- PC23.** ensure services are delivered equally to all clients regardless of personal and cultural beliefs
- PC24.** operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities
- PC25.** follow organisational guidelines and legal requirements on disclosure and confidentiality

*Uphold social diversity at the workplace*

To be competent, the user/individual on the job must be able to:

- PC26.** recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes
- PC27.** identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace
- PC28.** use inclusive or neutral language and gestures in all interactions
- PC29.** respect the personal and professional space of others
- PC30.** access grievance redressal mechanisms as per legislations

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organisation's policies on dress code, workplace timings, workplace behaviour, performance management, incentives, delivery standards, information security, etc.
- KU2.** organizational hierarchy and escalation matrix
- KU3.** importance of the individual's role in the workflow
- KU4.** organisational norms on health, safety and sustainability
- KU5.** work area inspection procedures and practices
- KU6.** professional etiquette and grooming

## Qualification Pack

- KU7.** communication etiquette across communicative mediums (online, digital, and in-person) including strategies/methods for sharing information, documentation, and providing and receiving feedback
- KU8.** importance of self-evaluations and developing a continuous learning and professional development plan
- KU9.** developments and trends impacting professional practice
- KU10.** importance of taking and using feedback from colleagues and clients to identify and introduce improvements in work performance
- KU11.** professional ethics and workplace norms on reporting and/or penalizing unethical behaviour and practices.
- KU12.** guidelines and legal requirements on disclosure, confidentiality, and conflicts of interest
- KU13.** strategies for collaboration with colleagues and clients.
- KU14.** professional responses and strategies against inappropriate language or behaviour toward self and others
- KU15.** Implicit bias (based on gender, disability, class, caste, colour, race, culture, religion, etc.) and its consequences in the workplace
- KU16.** organizational guidelines, prevalent legislations and accessibility norms and processes to support PwDs at the workplace
- KU17.** strategies for time, effort and resource allocation towards the goals.
- KU18.** basic concepts of work productivity including waste reduction, efficient material usage and optimization of time

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete documentation and forms such as work orders, invoices maintenance records activity logs, attendance sheets as per organizational format in English and/or local language
- GS2.** write basic accident or incident report accurately in an appropriate format
- GS3.** read warnings, instructions and other text material on product labels, components, etc. and relevant signages, warnings, labels or descriptions on equipment, etc. while carrying out work activities
- GS4.** convey and share technical information clearly using appropriate language
- GS5.** clarify task-related information
- GS6.** liaise with authorities and supervisors as per organizational protocol
- GS7.** listen, speak, and write in an inclusive, respectful manner in line with organizational protocol
- GS8.** seek clarification from immediate supervisor or responsible authority or exercise most appropriate solutions to safety breaches at work
- GS9.** report to the supervisor and when to deal with a colleague depending on the type of concern
- GS10.** deliver product to next work process on time
- GS11.** improve work process and report potential areas of delays and disruptions
- GS12.** communicate problems appropriately to others
- GS13.** identify symptoms of the fault to the cause of the problem and resolve, otherwise seek assistance and support from other sources to solve the problem

## Qualification Pack

- GS14.** anticipate and avoid hazards that may occur during repairs because of tools, materials used or repair processes
- GS15.** complete tasks efficiently and accurately within stipulated time
- GS16.** appreciate and respect social diversity in all professional settings
- GS17.** develop awareness and accountability for perspectives on gender, disabilities, and socio-cultural issues leading to discrimination, bias, or harassment at the workplace
- GS18.** maintain positive and effective relationships with colleagues and customers

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Communicate effectively at the workplace</i>                                                                                    | <b>5</b>     | <b>13</b>       | -             | -          |
| <b>PC1.</b> exchange information and instruction with colleagues, and seek clarifications and feedback as necessary                | 1            | 3               | -             | -          |
| <b>PC2.</b> assist colleagues where required                                                                                       | 1            | 3               | -             | -          |
| <b>PC3.</b> follow business communication etiquette in all interactions and communicative formats (online, digital, and in-person) | 1            | 4               | -             | -          |
| <b>PC4.</b> document and share all relevant information with stakeholders in agreed formats and as per agreed timelines            | 2            | 3               | -             | -          |
| <i>Work effectively</i>                                                                                                            | <b>6</b>     | <b>13</b>       | -             | -          |
| <b>PC5.</b> identify and obtain clarity regarding organisational, team and own goals and targets                                   | 1            | 2               | -             | -          |
| <b>PC6.</b> prioritise and plan work in order to achieve goals and targets                                                         | 1            | 2               | -             | -          |
| <b>PC7.</b> monitor own and team performance as per agreed plan                                                                    | 1            | 2               | -             | -          |
| <b>PC8.</b> complete duties accurately, systematically and within required timeframes                                              | 1            | 2               | -             | -          |
| <b>PC9.</b> express emotions appropriately at the workplace and manage own response to heightened emotions                         | 1            | 2               | -             | -          |
| <b>PC10.</b> maintain orderliness and cleanliness in the work area                                                                 | 1            | 3               | -             | -          |
| <i>Maintain and enhance professional competence</i>                                                                                | <b>8</b>     | <b>7</b>        | -             | -          |
| <b>PC11.</b> identify own strengths and weaknesses in relation to goals and targets                                                | 1            | 1               | -             | -          |
| <b>PC12.</b> adapt self, service, or product to meet success criteria                                                              | 1            | 1               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> seek and select opportunities for continuous professional development                                                                                                                                                                       | 1            | 1               | -             | -          |
| <b>PC14.</b> formulate a professional development plan to enhance capabilities                                                                                                                                                                           | 2            | 1               | -             | -          |
| <b>PC15.</b> build or contribute to the organizational knowledge base of cases, clients, issues, solutions, and innovations                                                                                                                              | 1            | 1               | -             | -          |
| <b>PC16.</b> examine developments and trends in field of work and their potential impact on work                                                                                                                                                         | 1            | 1               | -             | -          |
| <b>PC17.</b> take feedback from peers, supervisors and clients to improve own performance and practices                                                                                                                                                  | 1            | 1               | -             | -          |
| <i>Work in a disciplined and ethical manner</i>                                                                                                                                                                                                          | <b>11</b>    | <b>16</b>       | -             | -          |
| <b>PC18.</b> perform tasks as per workplace standards, organisational policies and legislative requirements                                                                                                                                              | 2            | 2               | -             | -          |
| <b>PC19.</b> display appropriate professional appearance at the workplace and adhere to the organisational dress code                                                                                                                                    | 1            | 2               | -             | -          |
| <b>PC20.</b> demonstrate responsible and disciplined behaviour at the workplace such as punctuality; completing tasks as per given time and standards; demonstrating professional behaviour at all times, adopting environment- friendly practices, etc. | 1            | 2               | -             | -          |
| <b>PC21.</b> identify the cause of conflict and options for resolution with peers or escalate grievances and problems to appropriate authority as per procedure for conflict resolution                                                                  | 2            | 2               | -             | -          |
| <b>PC22.</b> protect the rights of the client and organisation when delivering services                                                                                                                                                                  | 1            | 2               | -             | -          |
| <b>PC23.</b> ensure services are delivered equally to all clients regardless of personal and cultural beliefs                                                                                                                                            | 1            | 2               | -             | -          |
| <b>PC24.</b> operate within an agreed ethical code of practice and report unethical conduct to the appropriate authorities                                                                                                                               | 2            | 2               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> follow organisational guidelines and legal requirements on disclosure and confidentiality                                                            | 1            | 2               | -             | -          |
| <i>Uphold social diversity at the workplace</i>                                                                                                                   | <b>10</b>    | <b>11</b>       | -             | -          |
| <b>PC26.</b> recognize and evaluate biased practices against underrepresented groups like women and persons with disabilities, in workplace systems and processes | 2            | 2               | -             | -          |
| <b>PC27.</b> identify and report discrimination and harassment based on gender, disability, or cultural difference at the workplace                               | 2            | 2               | -             | -          |
| <b>PC28.</b> use inclusive or neutral language and gestures in all interactions                                                                                   | 2            | 2               | -             | -          |
| <b>PC29.</b> respect the personal and professional space of others                                                                                                | 2            | 2               | -             | -          |
| <b>PC30.</b> access grievance redressal mechanisms as per legislations                                                                                            | 2            | 3               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                  | <b>40</b>    | <b>60</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                    |
|----------------------------|------------------------------------|
| <b>NOS Code</b>            | ELE/N9905                          |
| <b>NOS Name</b>            | Work effectively at the workplace  |
| <b>Sector</b>              | Electronics                        |
| <b>Sub-Sector</b>          | Generic                            |
| <b>Occupation</b>          | Generic - Organizational Behaviour |
| <b>NSQF Level</b>          | 4                                  |
| <b>Credits</b>             | TBD                                |
| <b>Version</b>             | 2.0                                |
| <b>Last Reviewed Date</b>  | 24/02/2022                         |
| <b>Next Review Date</b>    | 30/12/2026                         |
| <b>NSQF Clearance Date</b> | 30/12/2021                         |

## Qualification Pack

### ELE/N1002: Apply health and safety practices at the workplace

#### Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace.

#### Scope

The scope covers the following :

- Deal with workplace hazards
- Apply fire safety practices
- Follow emergencies, rescue and first-aid procedures
- Effective waste management/recycling practices

#### Elements and Performance Criteria

##### *Deal with workplace hazards*

To be competent, the user/individual on the job must be able to:

- PC1.** identify job-site hazards and possible causes of accident in the workplace
- PC2.** perform work complying to organizational safe working practices and observing hazard signs displayed on containers, equipment and in various work areas such as inside buildings, in open areas and public spaces, etc.
- PC3.** use appropriate personal protective equipment (PPE) for specific tasks and work conditions, contaminant (concentration w.r.t air) requirements and severity of hazard while conforming to the Indian/International standards
- PC4.** follow standard safety procedures while handling tool/ ,equipment, hazardous substances and while working in hazardous environments
- PC5.** dispose electronic waste (such as toxins; metals such as lead, cadmium, barium; flame retardant plastics, welding slag etc.) as per industry approved techniques
- PC6.** avoid damage of components due to negligence in electrostatic discharge (ESD) procedures
- PC7.** locate general health and safety equipment in the workplace such as fire extinguishers; first aid equipment; safety instruments, clothing and installations (fire exits, exhaust fans)
- PC8.** maintain appropriate posture while handling heavy objects
- PC9.** apply good housekeeping practices at all times

##### *Apply fire safety practices*

To be competent, the user/individual on the job must be able to:

- PC10.** take preventive measures to prevent fire hazards
- PC11.**
  - use appropriate fire extinguishers for different types of fires
  - Types of fires: Class A: e.g. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: e.g. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no I
- PC12.** exhibit rescue and first-aid techniques in case of fire or electrocution

## Qualification Pack

### *Follow emergencies, rescue and first-aid procedures*

To be competent, the user/individual on the job must be able to:

- PC13.** administer appropriate first aid to victims in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC14.** administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock,
- PC15.** participate regularly in emergency procedures such as raising alarm, safe/efficient, evacuation, correct means of taking shelter and escaping, correct assembly point, roll call, correct return to work
- PC16.** use correct method to move injured people and others during an emergency

### *Effective waste management/recycling practices*

To be competent, the user/individual on the job must be able to:

- PC17.** identify recyclable and non-recyclable, and hazardous waste generated
- PC18.** segregate waste into different categories
- PC19.** ensure disposal of non-recyclable waste appropriately
- PC20.** deposit non-recyclable and reusable material at identified location
- PC21.** follow processes specified for disposal of hazardous waste

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of working in clean and safe work environment following safety practices and procedures
- KU2.** health and safety roles and responsibilities of relevant personnel within and outside the organisation
- KU3.** key internal and external sources of health and safety information
- KU4.** basic knowledge of electronic devices and related health risks
- KU5.** meaning of hazards and risks
- KU6.** various types of health and safety hazards commonly present in the work environment such as physical hazards, electrical hazards, chemical hazards, fire hazards, equipment related hazards, health hazards, etc.
- KU7.** methods of accident prevention
- KU8.** importance of using protective clothing/equipment while working
- KU9.** general principles for identifying and controlling health and safety risks
- KU10.** main hazards and preventive as well as control measures while working with different types of equipment
- KU11.** importance of carrying out electrical and non-electrical isolation to prevent hazards from loss of machine/system/process control
- KU12.** main hazards and preventive as well as control measures when working with electrical systems or using electrical equipment
- KU13.** forms and classifications of hazardous substances
- KU14.** safe working practices while working at various hazardous sites
- KU15.** prevention and control measures to reduce risks from exposure to hazardous substances

## Qualification Pack

- KU16.** health effects associated with exposure to noise and vibration and the appropriate control measures
- KU17.** precautionary activities to prevent the fire accident
- KU18.** various causes of fire such as heating of metal, spontaneous ignition, sparking, electrical eating, loose fires (smoking, welding, etc.) chemical fires etc.
- KU19.** techniques of using the different fire extinguishers
- KU20.** different methods and material to extinguish fires
- KU21.** different materials used for extinguishing fire such as sand, water, foam, CO<sub>2</sub>, dry powder
- KU22.** rescue techniques used during a fire hazard
- KU23.** various types of safety signs and their meaning
- KU24.** basic first aid treatment relevant to the common work place injuries e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU25.** contents of written accident report
- KU26.** potential injuries and ill health associated with incorrect handling of tools and equipment
- KU27.** safe lifting and carrying practices
- KU28.** potential impact to a person who is moved incorrectly
- KU29.** personal safety, health and dignity issues relating to the movement of a person by others
- KU30.** ESD measures and 5S
- KU31.** efficient utilization and management of material and water
- KU32.** ways to recognize common electrical problems and practices of conserving electricity
- KU33.** usage of different colours of dustbins, categorization of waste into dry, wet, recyclable, nonrecyclable and items of single-use plastics
- KU34.** organization's procedure for minimizing waste
- KU35.** waste management and methods of waste disposal
- KU36.** common sources of pollution and ways to minimize it
- KU37.** names, contact information and location of people responsible for health and safety in the workplace
- KU38.** location of documents and equipment for health and safety compliance/practices in the workplace
- KU39.** safety notices, signs and instructions at workplace

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret general health and safety guidelines labels, charts, signages
- GS2.** read operation manuals
- GS3.** write health and safety compliance report
- GS4.** write an accident/incident report in local language or English
- GS5.** provide an emergency or safety incident brief to seniors or relevant authorities in a calm, clear and to-the-point manner
- GS6.** communicate general health and safety guidelines to colleagues/co-workers

## Qualification Pack

- GS7.** communicate appropriately with co-workers in order to clarify instructions and other issues
- GS8.** act in case of any potential hazards observed in the work place
- GS9.** plan and organize their own work schedule, work area, tools, equipment in compliance with organizational policies for health, safety and security
- GS10.** take adequate measures to ensure the safety of clients and visitors at the workplace
- GS11.** identify immediate or temporary solutions to resolve delays
- GS12.** evaluate the work area for health and safety risks or hazards
- GS13.** use cause and effect relations to anticipate potential issues, problems and their solution in the work area related to safety
- GS14.** recognise emergency and potential emergency situations
- GS15.** protect self and others from a health and safety risk or hazard
- GS16.** communicate and collaborate to incorporate sustainable practices (greening) in workplace processes
- GS17.** record data on waste disposal at workplace

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Deal with workplace hazards</i>                                                                                                                                                                                                       | <b>20</b>    | <b>31</b>       | -             | -          |
| <b>PC1.</b> identify job-site hazards and possible causes of accident in the workplace                                                                                                                                                   | 2            | 3               | -             | -          |
| <b>PC2.</b> perform work complying to organizational safe working practices and observing hazard signs displayed on containers, equipment and in various work areas such as inside buildings, in open areas and public spaces, etc.      | 3            | 4               | -             | -          |
| <b>PC3.</b> use appropriate personal protective equipment (PPE) for specific tasks and work conditions, contaminant (concentration w.r.t air) requirements and severity of hazard while conforming to the Indian/International standards | 3            | 4               | -             | -          |
| <b>PC4.</b> follow standard safety procedures while handling tool/ ,equipment, hazardous substances and while working in hazardous environments                                                                                          | 3            | 4               | -             | -          |
| <b>PC5.</b> dispose electronic waste (such as toxins; metals such as lead, cadmium, barium; flame retardant plastics, welding slag etc.) as per industry approved techniques                                                             | 2            | 4               | -             | -          |
| <b>PC6.</b> avoid damage of components due to negligence in electrostatic discharge (ESD) procedures                                                                                                                                     | 2            | 3               | -             | -          |
| <b>PC7.</b> locate general health and safety equipment in the workplace such as fire extinguishers; first aid equipment; safety instruments, clothing and installations (fire exits, exhaust fans)                                       | 2            | 3               | -             | -          |
| <b>PC8.</b> maintain appropriate posture while handling heavy objects                                                                                                                                                                    | 1            | 3               | -             | -          |
| <b>PC9.</b> apply good housekeeping practices at all times                                                                                                                                                                               | 2            | 3               | -             | -          |
| <i>Apply fire safety practices</i>                                                                                                                                                                                                       | <b>4</b>     | <b>9</b>        | -             | -          |
| <b>PC10.</b> take preventive measures to prevent fire hazards                                                                                                                                                                            | 2            | 3               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> <ul style="list-style-type: none"> <li>use appropriate fire extinguishers for different types of fires</li> <li>Types of fires: Class A: e.g. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: e.g. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no I</li> </ul> | 1            | 3               | -             | -          |
| <b>PC12.</b> exhibit rescue and first-aid techniques in case of fire or electrocution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1            | 3               | -             | -          |
| <i>Follow emergencies, rescue and first-aid procedures</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>6</b>     | <b>13</b>       | -             | -          |
| <b>PC13.</b> administer appropriate first aid to victims in case of bleeding, burns, choking, electric shock, poisoning etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1            | 3               | -             | -          |
| <b>PC14.</b> administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock,                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1            | 2               | -             | -          |
| <b>PC15.</b> participate regularly in emergency procedures such as raising alarm, safe/efficient, evacuation, correct means of taking shelter and escaping, correct assembly point, roll call, correct return to work                                                                                                                                                                                                                                                                                                                                                                       | 2            | 4               | -             | -          |
| <b>PC16.</b> use correct method to move injured people and others during an emergency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2            | 4               | -             | -          |
| <i>Effective waste management/recycling practices</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>5</b>     | <b>12</b>       | -             | -          |
| <b>PC17.</b> identify recyclable and non-recyclable, and hazardous waste generated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1            | 3               | -             | -          |
| <b>PC18.</b> segregate waste into different categories                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1            | 2               | -             | -          |
| <b>PC19.</b> ensure disposal of non-recyclable waste appropriately                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1            | 2               | -             | -          |
| <b>PC20.</b> deposit non-recyclable and reusable material at identified location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1            | 3               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC21.</b> follow processes specified for disposal of hazardous waste | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                        | <b>35</b>    | <b>65</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>NOS Code</b>            | ELE/N1002                                          |
| <b>NOS Name</b>            | Apply health and safety practices at the workplace |
| <b>Sector</b>              | Electronics                                        |
| <b>Sub-Sector</b>          | Generic                                            |
| <b>Occupation</b>          | Generic - Health Safety                            |
| <b>NSQF Level</b>          | 4                                                  |
| <b>Credits</b>             | TBD                                                |
| <b>Version</b>             | 3.0                                                |
| <b>Last Reviewed Date</b>  | 24/02/2022                                         |
| <b>Next Review Date</b>    | 24/02/2025                                         |
| <b>NSQC Clearance Date</b> | 24/02/2022                                         |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings

## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |



### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Qualification Pack

### ELE/N8120: Lay, install and carry out commissioning for multi play network

#### Description

This OS unit about carrying out laying, installing and commissioning of multi play network

#### Scope

The scope covers the following :

- Lay optical fibre cables for multi play network
- Carry out commissioning and installation of multi play network

#### Elements and Performance Criteria

##### *Lay optical fibre cables for multi play network*

To be competent, the user/individual on the job must be able to:

- PC1.** check the feasibility of laying trunk and sub-trunk routes from the fibre to the home (FTTH) distribution network plan received from route designer
- PC2.** create a requirement list and carry out procurement of the materials
- PC3.** install network management system (NMS) modem at each node to ensure the node is functional 24x7
- PC4.** ensure to carry the right of way (ROW) letter given by the concerned authority for laying the cables
- PC5.** lay the cables as per the requirement plan using proper messenger and clipping wires
- PC6.** prepare as build diagram (ABD) of the complete fibre trunk route, poles used, locations of joint closures, splitters and other passives by marking on a google map with all physical locations
- PC7.** ensure a loop of 2-3 meters are maintained on every alternative poles
- PC8.** check a proper bend radius is maintained as per manufacturer's specifications
- PC9.** use all the safety measures related to work
- PC10.** maintain a minimum height of 12 feet for the trunk line
- PC11.** install passive splitters as per design diagram and requirement at multiple locations on the trunk line
- PC12.** ensure proper fitment of joint closure on the pole using the clamp
- PC13.** use drip loop of fibre cable at the input and output of joint closure
- PC14.** ensure to maintain a split ratio for creating customer ports from one fibre Split ratio: 1:8, 1:16, 1:24, 1:32, etc.
- PC15.** prepare the fibre cables for jointing as per the colour code
- PC16.** carry out work in a moisture free environment
- PC17.** ensure the fibre cables are spliced as per the design diagram by optical fibre splicer
- PC18.** clean dust particles on the fibre by using disinfectant solution, e.g., iso-propyl solution

## Qualification Pack

**PC19.** splice the fibre cable by maintaining the standard splicing parameters like attenuation in dB per splice

**PC20.** use proper tools for fibre cable laying and splicing Tools: Plier, spanner, screw driver, line tester, cleaver, splicing machine

### *Carry out commissioning and installation of multi play network*

To be competent, the user/individual on the job must be able to:

**PC21.** carry out testing of network by energising the fibre core at the backend using a laser source

**PC22.** test the power at various splitter output points using an optical power meter

**PC23.** ensure the available powers at all the splitter output points are as per standard for driving the customer premise equipment (CPE)

**PC24.** check the feed feasibility and amount of drop cable required as per customers request

**PC25.** carry all the materials like drop cable, joint closure, modem, patch cord, multi play ready optical network unit (ONU), LAN cable, RF cable, set top box (STB), and high definition multiple insertion (HDMI) cable

**PC26.** connect the drop cable to the nearest available splitter port of the joint closure in the house

**PC27.** ensure the output of the joint closure is fed to the optical network unit using a patch cord

**PC28.** carry out signal level testing using an optical power meter at the end of the patch cord as per standard to drive the customer premise equipment

**PC29.** connect the radio frequency (RF) to the set top box for establishing signal for television network

**PC30.** use the LAN cable to connect and configure the WIFI router

**PC31.** ensure the WIFI router is authorised as per the plan opted by the customer through customer provisioning system at the backend

**PC32.** escalate customer issues and problems that are unresolved at field level

**PC33.** record all the optical parameters, RF parameters for forward and reverse path for final audit and closure of the activity

**PC34.** install and configure the modem, and connect it to the laptops/ desktops

**PC35.** test the signal levels and other technical parameters required for cable modem

**PC36.** describe the operations and functions of cable modem to the customers

**PC37.** explain the preventive maintenance procedures to keep the connection up and running

**PC38.** clean the work site after the work is complete, and remove all the wastes from the site

**PC39.** ensure the customer signs off the installation report along with the customer feedback form

**PC40.** update installation status to the back office after satisfactory closure of the work order

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** companys policies on: Incentives, delivery standards and personnel management and customer service standards

**KU2.** reporting and documentation processes

**KU3.** reporting structure

**KU4.** optical fibre cables and its types

## Qualification Pack

- KU5.** fibre to the home (FTTH) distribution network
- KU6.** network management system (NMS) modem
- KU7.** right of way (ROW) letter
- KU8.** messenger and clipping wires
- KU9.** as build diagram (ABD) of the complete fibre trunk route, poles used, locations of joint closures, splitters and other passives
- KU10.** bend radius
- KU11.** safety measures
- KU12.** procedures for jointing of optical fibre cables
- KU13.** splicing method and its standard parameters
- KU14.** proper tools for optical fibre cable laying and splicing
- KU15.** network testing using laser source
- KU16.** splitters and its functions
- KU17.** optical power meter and its uses
- KU18.** operations and functions of various parts of multi play network: Various parts: Drop cable, joint closure, patch cord, modem, multi play ready optical network unit (ONU), LAN cable, RF cable, Set top box (STB), High definition multiple insertion (HDMI) cable, etc
- KU19.** process for signal level testing
- KU20.** optical parameters and RF parameters for forward and reverse path
- KU21.** installation and configuration process of modem and set top box

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete forms such as work orders, maintenance records, customer feedback form, etc.
- GS2.** note problems on job sheet and details of work done
- GS3.** read warnings, instructions and other text material on product labels, components, etc
- GS4.** read job sheets or work orders
- GS5.** read product and model serial numbers and interpret details such as make date, availability, etc
- GS6.** receive and ask for clarifications from supervisor on the job requirement
- GS7.** listen carefully to the customer
- GS8.** communicate in local/English language
- GS9.** educate and inform customer about product, warranty, cost of service, model replacement, different plans for subscription, etc
- GS10.** educate on precautions to be taken in order to avoid recurrence of problem
- GS11.** follow standard operating procedures while making decisions
- GS12.** take approval from supervisor in case the decision has to be made for exceptions
- GS13.** work with supervisor and co-workers to achieve smooth workflow
- GS14.** work with superiors and co-workers to share knowledge and learning
- GS15.** maintain personal grooming

## Qualification Pack

- GS16.** be polite, patient and courteous under all circumstances with all types of customers
- GS17.** decide on the spot on whether interaction of customer with superior is necessary or not
- GS18.** maintain proper etiquette including the appropriate physical distance with customer during conversation, not entering customer premises without permission
- GS19.** seek inputs at assessing the problems
- GS20.** interpret accurately drawings, wiring and job specifications/instructions
- GS21.** improve work processes

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Lay optical fibre cables for multi play network</i>                                                                                                                                                     | <b>17</b>    | <b>37</b>       | -             | -          |
| <b>PC1.</b> check the feasibility of laying trunk and sub-trunk routes from the fibre to the home (FTTH) distribution network plan received from route designer                                            | 1            | 2               | -             | -          |
| <b>PC2.</b> create a requirement list and carry out procurement of the materials                                                                                                                           | 1            | 2               | -             | -          |
| <b>PC3.</b> install network management system (NMS) modem at each node to ensure the node is functional 24x7                                                                                               | 1            | 2               | -             | -          |
| <b>PC4.</b> ensure to carry the right of way (ROW) letter given by the concerned authority for laying the cables                                                                                           | 1            | 2               | -             | -          |
| <b>PC5.</b> lay the cables as per the requirement plan using proper messenger and clipping wires                                                                                                           | 1            | 2               | -             | -          |
| <b>PC6.</b> prepare as build diagram (ABD) of the complete fibre trunk route, poles used, locations of joint closures, splitters and other passives by marking on a google map with all physical locations | 1            | 2               | -             | -          |
| <b>PC7.</b> ensure a loop of 2-3 meters are maintained on every alternative poles                                                                                                                          | -            | 1               | -             | -          |
| <b>PC8.</b> check a proper bend radius is maintained as per manufacturer's specifications                                                                                                                  | -            | 2               | -             | -          |
| <b>PC9.</b> use all the safety measures related to work                                                                                                                                                    | 1            | 1               | -             | -          |
| <b>PC10.</b> maintain a minimum height of 12 feet for the trunk line                                                                                                                                       | 1            | 2               | -             | -          |
| <b>PC11.</b> install passive splitters as per design diagram and requirement at multiple locations on the trunk line                                                                                       | 1            | 2               | -             | -          |
| <b>PC12.</b> ensure proper fitment of joint closure on the pole using the clamp                                                                                                                            | 1            | 2               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> use drip loop of fibre cable at the input and output of joint closure                                                                                                                                               | 1            | 2               | -             | -          |
| <b>PC14.</b> ensure to maintain a split ratio for creating customer ports from one fibre Split ratio: 1:8, 1:16, 1:24, 1:32, etc.                                                                                                | 1            | 2               | -             | -          |
| <b>PC15.</b> prepare the fibre cables for jointing as per the colour code                                                                                                                                                        | 1            | 2               | -             | -          |
| <b>PC16.</b> carry out work in a moisture free environment                                                                                                                                                                       | -            | 1               | -             | -          |
| <b>PC17.</b> ensure the fibre cables are spliced as per the design diagram by optical fibre splicer                                                                                                                              | 1            | 2               | -             | -          |
| <b>PC18.</b> clean dust particles on the fibre by using disinfectant solution, e.g., iso-propyl solution                                                                                                                         | 1            | 2               | -             | -          |
| <b>PC19.</b> splice the fibre cable by maintaining the standard splicing parameters like attenuation in dB per splice                                                                                                            | 1            | 2               | -             | -          |
| <b>PC20.</b> use proper tools for fibre cable laying and splicing Tools: Plier, spanner, screw driver, line tester, cleaver, splicing machine                                                                                    | 1            | 2               | -             | -          |
| <i>Carry out commissioning and installation of multi play network</i>                                                                                                                                                            | <b>13</b>    | <b>33</b>       | -             | -          |
| <b>PC21.</b> carry out testing of network by energising the fibre core at the backend using a laser source                                                                                                                       | 1            | 2               | -             | -          |
| <b>PC22.</b> test the power at various splitter output points using an optical power meter                                                                                                                                       | 1            | 2               | -             | -          |
| <b>PC23.</b> ensure the available powers at all the splitter output points are as per standard for driving the customer premise equipment (CPE)                                                                                  | 1            | 2               | -             | -          |
| <b>PC24.</b> check the feed feasibility and amount of drop cable required as per customers request                                                                                                                               | 1            | 2               | -             | -          |
| <b>PC25.</b> carry all the materials like drop cable, joint closure, modem, patch cord, multi play ready optical network unit (ONU), LAN cable, RF cable, set top box (STB), and high definition multiple insertion (HDMI) cable | 1            | 2               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC26.</b> connect the drop cable to the nearest available splitter port of the joint closure in the house                                                  | 1            | 2               | -             | -          |
| <b>PC27.</b> ensure the output of the joint closure is fed to the optical network unit using a patch cord                                                     | 1            | 2               | -             | -          |
| <b>PC28.</b> carry out signal level testing using an optical power meter at the end of the patch cord as per standard to drive the customer premise equipment | 1            | 2               | -             | -          |
| <b>PC29.</b> connect the radio frequency (RF) to the set top box for establishing signal for television network                                               | 1            | 2               | -             | -          |
| <b>PC30.</b> use the LAN cable to connect and configure the WIFI router                                                                                       | 1            | 2               | -             | -          |
| <b>PC31.</b> ensure the WIFI router is authorised as per the plan opted by the customer through customer provisioning system at the backend                   | 1            | 2               | -             | -          |
| <b>PC32.</b> escalate customer issues and problems that are unresolved at field level                                                                         | 1            | 2               | -             | -          |
| <b>PC33.</b> record all the optical parameters, RF parameters for forward and reverse path for final audit and closure of the activity                        | 1            | 2               | -             | -          |
| <b>PC34.</b> install and configure the modem, and connect it to the laptops/ desktops                                                                         | -            | 1               | -             | -          |
| <b>PC35.</b> test the signal levels and other technical parameters required for cable modem                                                                   | -            | 1               | -             | -          |
| <b>PC36.</b> describe the operations and functions of cable modem to the customers                                                                            | -            | 1               | -             | -          |
| <b>PC37.</b> explain the preventive maintenance procedures to keep the connection up and running                                                              | -            | 1               | -             | -          |
| <b>PC38.</b> clean the work site after the work is complete, and remove all the wastes from the site                                                          | -            | 1               | -             | -          |
| <b>PC39.</b> ensure the customer signs off the installation report along with the customer feedback form                                                      | -            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC40.</b> update installation status to the back office after satisfactory closure of the work order | -            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                        | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                 |
|----------------------------|-----------------------------------------------------------------|
| <b>NOS Code</b>            | ELE/N8120                                                       |
| <b>NOS Name</b>            | Lay, install and carry out commissioning for multi play network |
| <b>Sector</b>              | Electronics                                                     |
| <b>Sub-Sector</b>          | Communication and Broadcasting                                  |
| <b>Occupation</b>          | After Sales Service-C&B                                         |
| <b>NSQF Level</b>          | 4                                                               |
| <b>Credits</b>             | 2                                                               |
| <b>Version</b>             | 2.0                                                             |
| <b>Last Reviewed Date</b>  | NA                                                              |
| <b>Next Review Date</b>    | 27/01/2027                                                      |
| <b>NSQC Clearance Date</b> | 27/01/2022                                                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

## Qualification Pack

**Minimum Aggregate Passing % at QP Level : 70**

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ELE/N8116.Lay truck or sub truck cables install and connectorize as per designed route        | 40           | 60              | 0             | 0          | 100         | 15         |
| ELE/N8117.Carry out installation of power inserter & balancing and commissioning of amplifier | 40           | 60              | 0             | 0          | 100         | 20         |
| ELE/N8118.Install Set Top Box at customers premise                                            | 40           | 60              | 0             | 0          | 100         | 10         |
| ELE/N8119.Cater to customer complaints for troubleshooting and maintenance of set top box     | 30           | 70              | 0             | 0          | 100         | 15         |
| ELE/N9905.Work effectively at the workplace                                                   | 40           | 60              | -             | -          | 100         | 10         |
| ELE/N1002.Apply health and safety practices at the workplace                                  | 35           | 65              | -             | -          | 100         | 10         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                                                 | 20           | 30              | 0             | 0          | 50          | 10         |
| ELE/N8120.Lay, install and carry out commissioning for multi play network                     | 30           | 70              | -             | -          | 100         | 10         |
| <b>Total</b>                                                                                  | <b>275</b>   | <b>475</b>      | <b>0</b>      | <b>0</b>   | <b>750</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |